

Volatility Forecasting with Residual Learning

Eddie Liu | GARCH, LSTM, and empirical model evaluation

An additive LSTM reduces observed forecast MSE by **5.48%** versus GARCH with a fixed floor. But the uncertainty interval includes zero, and invalid raw variance forecasts remain a material limitation.

3,738

common test dates

178

walk-forward folds

2010-2024

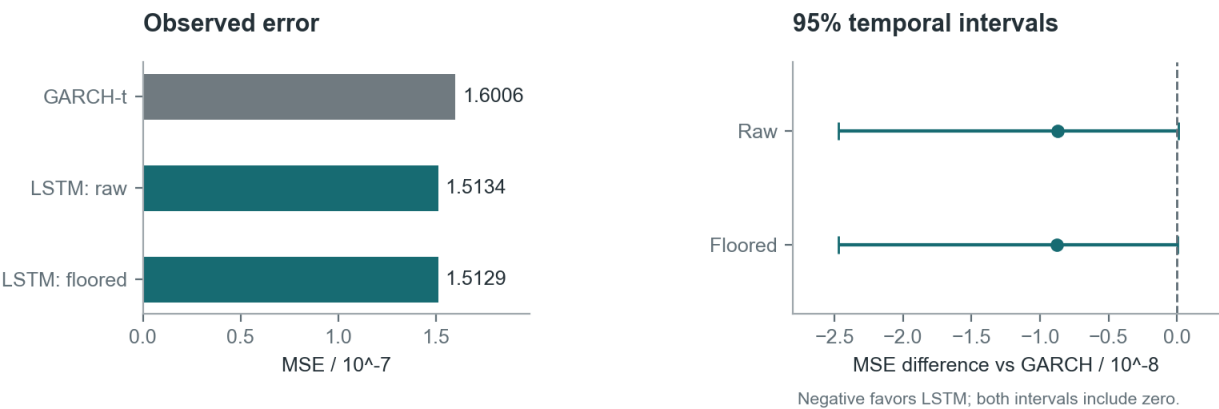
historical evaluation period

Can an LSTM learn GARCH's forecast errors?

The target is one-day S&P 500 variance, measured with squared daily log returns.

01 Forecast the baseline	02 Learn its errors	03 Evaluate both outputs
GARCH-t estimates next-day variance g using past returns.	LSTM predicts the signed error $e = y - g$ from lagged features.	Raw $h = g + \text{predicted } e$. Floored $h = \max(\text{raw } h, 1e-12)$.

The evidence at a glance



Paired stationary bootstrap: mean block length 21; 2,000 resamples. Intervals condition on the trained forecasts, not training-seed or model-selection uncertainty.

Raw MSE improves 5.45%, versus 5.48% when floored: the gain is not primarily a clipping effect.

Validity: 60 raw forecasts are nonpositive. Floored QLIKE is 457,801, versus -8.4613 for GARCH; lower is better. Lower MSE alone is not a deployable-model result.

What the experiment establishes

Controlled historical evaluation

Sample	2010-02-05 to 2024-12-11; cached ^GSPC prices from 2003-2024. One index; one training-seed schedule.
Timing	21 lagged input rows ending at t-1. Training-only scaling; expanding training history, 252 validation observations, and 21-day non-overlapping test blocks.
Model family	GARCH-t, EWMA, historical variance, six log-target/log-ratio LSTM/GRU specifications, and one signed-residual LSTM. Eight recurrent units; validation-based early stopping.

Same-date forecast comparison

Model / family	MSE / 10 ⁻⁷	QLIKE
GARCH-t	1.6006	-8.4613
EWMA 0.94	1.7482	-8.4200
Lagged RV21	1.8721	-8.3529
Log-target / log-ratio RNNs (6)	1.8513 to 1.9701	-6.6595 to -6.4303
Additive LSTM: raw	1.5134	Undefined
Additive LSTM: floored	1.5129	457,801.1

Lower is better for both losses. The RNN row gives separate metric ranges across six specifications, not one model. Raw and floored additive rows use the same fitted LSTM. Raw QLIKE is undefined for nonpositive variance forecasts.

Three research takeaways

Objective alignment matters. Signed-residual MSE matches raw variance-forecast MSE. Log-target MSE is different; all six log-target/log-ratio candidates lose to GARCH on both reported losses.

Error and validity are separate. A floor does not solve positivity or calibration. The primary MSE interval includes zero, with significance sensitive to block length.

The workflow is inspectable. Python code includes chronological splits, daily GARCH updates, matched-date scoring, source hashes, tests, and a synthetic offline demo.

Scope and next experiment

The additive model was tested after inspecting this historical sample, not on an untouched holdout. No trading alpha or production validation is claimed. Proposed next test: positive, mean-targeted forecasts across multiple seeds and new data; not yet run.

Full report (6 pages) | Code, demo, and evidence

Sources: reports/corrected_results.json and reports/additive_results.json. Full raw market snapshots and predictions are not bundled. Based on a collaborative research project; this brief is prepared by Eddie Liu.